

Predictive Modeling for Click-Through Rate Optimization at ConnectSphere Digital

A logistic regression approach to identifying the users most likely to engage with online advertisements.

Client	ConnectSphere Digital
Domain	Digital advertising analytics
Task	Binary classification (click / no click)
Model	Logistic Regression (scikit-learn)
Dataset	advertising.csv, 1,000 users, 10 attributes

Contents

- | | |
|--|--|
| 1. Executive summary | 7. Model development |
| 2. Problem statement and objectives | 8. Results and evaluation |
| 3. Dataset description | 9. Drivers of ad clicks |
| 4. Data quality assessment | 10. Business implications |
| 5. Exploratory data analysis | 11. Limitations and next steps |
| 6. Feature engineering and preparation | 12. Conclusion |

1. Executive summary

ConnectSphere Digital runs advertising campaigns for many clients, but a large share of ad spend reaches users who are unlikely to engage. This project builds a logistic regression model that estimates, for each user, the probability of clicking an ad, so that budget can be directed toward the users who matter most.

The model was trained on 800 users and tested on 200 unseen users. It classified 98% of the test users correctly, and its click predictions were right 99% of the time. The strongest signals were behavioral: users who spend less

time on the site and less time online each day are far more likely to click, and older users click more than younger ones.

98.0%

Accuracy

98.98%

Precision

97.0%

Recall

0.9915

ROC-AUC

2. Problem statement and objectives

Problem

Much of the advertising budget is spent showing ads to internet users with a low probability of engagement. This untargeted approach weakens campaign performance and lowers Return on Ad Spend (ROAS) for clients. The agency has no systematic, data-driven method for identifying and prioritizing users who are likely to interact with an ad.

Project goal

Develop and evaluate a logistic regression model that predicts whether a user will click an online ad, using demographic data (Age, Area Income) and behavioral data (Daily Time Spent on Site, Daily Internet Usage). Users are classified as *likely to click* (1) or *unlikely to click* (0), and reliability is judged by accuracy, precision and recall.

Business objective

Use the model to optimize ad targeting, allocate budgets toward higher-propensity segments, and raise the overall Click-Through Rate (CTR) across campaigns. The expected outcome is better client satisfaction and a stronger competitive position.

3. Dataset description

The dataset holds 1,000 historical records of users exposed to online ads. Each record combines demographics, engagement behavior, location and ad details, and the final outcome.

Variable	Type	Description
Daily Time Spent on Site	Numeric	Minutes the user spends on the site each day
Age	Integer	Age of the user in years
Area Income	Numeric	Average income of the user's geographic area
Daily Internet Usage	Numeric	Minutes the user spends online each day
Male	Binary	1 if male, 0 otherwise
Ad Topic Line, City, Country	Text	Ad headline and user location
Timestamp	Datetime	Time the ad was shown
Clicked on Ad	Binary	Target: 1 if the user clicked, 0 otherwise

Feature	Mean	Std	Min	Median	Max
Daily Time Spent on Site	65.00	15.85	32.60	68.22	91.43
Age	36.01	8.79	19	35	61
Area Income	55,000	13,415	13,997	57,012	79,485
Daily Internet Usage	180.00	43.90	104.78	183.13	269.96

Table 1. Summary statistics of numeric features (n = 1,000).

4. Data quality assessment

- **Missing values:** none in any of the 10 columns (0.0%), so no imputation or row removal was needed.
- **Duplicates:** 0 duplicate rows; every observation is unique.
- **Class balance:** exactly 500 clickers and 500 non-clickers (50% each), so accuracy is a fair headline metric and no resampling was required.
- **Outliers (IQR method):** only Area Income showed outliers, 9 low-income observations below the lower bound of about 19,374. Time on site, age and internet usage had none. These records look like genuine low-income areas rather than errors, so they were retained.

5. Exploratory data analysis

Feature behavior by outcome

Comparing clickers with non-clickers revealed clear separation on all four numeric features. Non-clickers typically spend more time on the site, are more active online, live in higher-income areas and are younger (median age about 31 versus 40 for clickers).

Feature	Correlation	Direction
Daily Internet Usage	-0.79	Higher usage, fewer clicks
Daily Time Spent on Site	-0.75	More time on site, fewer clicks
Age	+0.49	Older users click more
Area Income	-0.48	Higher-income areas, fewer clicks

Table 2. Correlation of each numeric feature with the target (Clicked on Ad).

Time on site and internet usage are also positively correlated with each other (0.52), so some of their information overlaps. Logistic regression handles this level of correlation well, but it is worth remembering when comparing coefficients.

Temporal patterns

The timestamp was decomposed into hour, month and day of week. Click rate ranged from roughly 40% to 61% across hours and from about 45% to 55% across weekdays, with Thursday highest and Tuesday lowest. The hourly pattern is jagged with no smooth daily rhythm, which suggests that much of this variation is noise from small samples (about 40 users per hour). The model later confirmed that these time features carry little signal.

6. Feature engineering and preparation

- **Timestamp features:** Hour, Month and DayOfWeek were extracted and the raw timestamp removed.
- **Dropped columns:** Ad Topic Line, City and Country were excluded. They are high-cardinality text fields that would need an explicit encoding strategy, so the baseline model uses a focused numeric feature set.
- **Final feature set (8):** Daily Time Spent on Site, Age, Area Income, Daily Internet Usage, Male, Hour, Month, DayOfWeek.
- **Train-test split:** 80/20 (800 training, 200 testing), stratified on the target and seeded with `random_state=42` for reproducibility.

- **Scaling:** `StandardScaler` was fitted on the training set only and then applied to the test set, which prevents data leakage and makes coefficients comparable.

7. Model development

Logistic regression was chosen because the target is binary, the model outputs a click probability that can be used to rank users, and its coefficients are directly interpretable, which matters when explaining targeting decisions to clients. The model was trained with `max_iter=1000` and `random_state=42`, and predictions were made on the held-out test set at the default 0.5 threshold.

8. Results and evaluation

Metric	Score	Meaning
Accuracy	0.9800	Share of all users classified correctly
Precision	0.9898	Of users predicted to click, the share who did
Recall	0.9700	Of actual clickers, the share the model found
F1-score	0.9798	Balance of precision and recall
ROC-AUC	0.9915	Ability to rank clickers above non-clickers
Average precision	0.9935	Area under the precision-recall curve

Table 3. Test-set performance (200 users).

	Predicted: No click	Predicted: Click
Actual: No click	99 true negatives	1 false positives
Actual: Click	3 false negatives	97 true positives

Figure 1. Confusion matrix on the test set. Only 4 of 200 users were misclassified.

The model made just one false-positive error (a user predicted to click who did not) and three false-negative errors (clickers it missed). Class-level F1 was 0.98 for both classes, and the predicted class split (roughly 102

versus 98) closely matched the true 100/100 split, so the model shows no meaningful bias toward either class. The ROC curve rises almost vertically and the precision-recall curve stays near 1.0 until recall passes about 0.9, indicating strong separation at almost any threshold.

9. Drivers of ad clicks

Because features were standardized, coefficient sizes are comparable. Negative values reduce click odds, and positive values increase them.

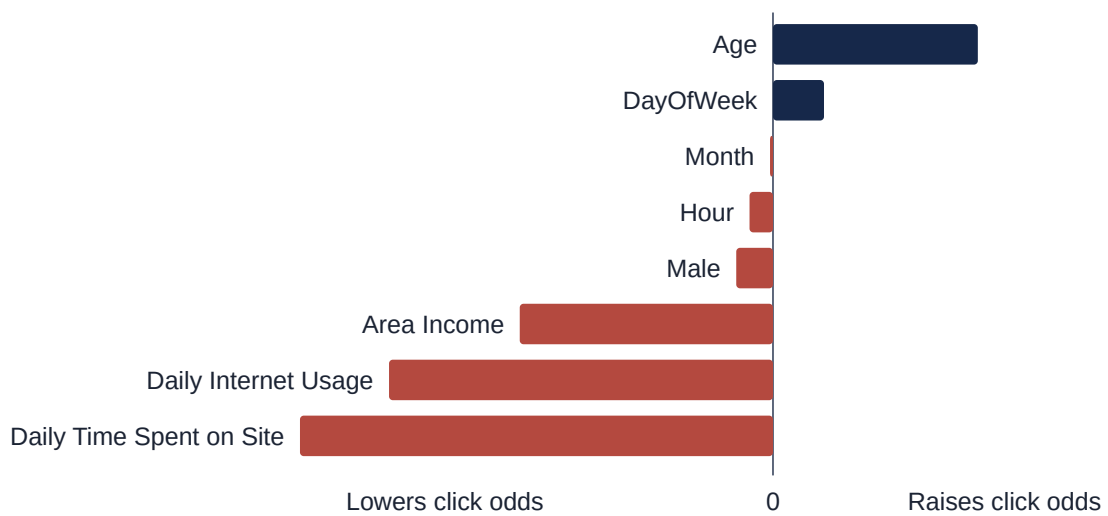


Figure 2. Standardized logistic regression coefficients.

Feature	Coefficient	Odds ratio	Interpretation
Age	+1.249	3.49	About 8.8 years older multiplies click odds by 3.5
DayOfWeek	+0.311	1.36	Minor effect
Month	-0.018	0.98	No meaningful effect
Hour	-0.143	0.87	Minor effect
Male	-0.224	0.80	Small effect
Area Income	-1.544	0.21	About \$13,400 higher income cuts click odds by 79%
Daily Internet Usage	-2.341	0.10	About 44 more minutes online cuts click odds by 90%
Daily Time Spent on Site	-2.884	0.06	About 16 more minutes on site cuts click odds by 94%

Table 4. Odds ratios per one standard-deviation increase, other features held constant.

Daily Time Spent on Site, Daily Internet Usage, Area Income and Age are the four features that drive the model. The time-based features and gender contribute very little, which supports the decision to keep the model simple.

Reading the pattern. Heavy site users and heavy internet users click *less*. A plausible explanation is that highly engaged users are more familiar with ads and more likely to ignore them, while lighter users respond more readily. This is an association in the data, not proof of cause, and it should be validated on the client's own campaigns.

10. Business implications

- **Prioritize by probability.** Score every user with the model and concentrate spend on the highest click probabilities instead of treating all users equally.
- **Build audience segments.** Older users with lower daily internet usage, lower time on site and lower area income form the highest-propensity segment.

- **Tune the threshold to the cost.** Use a lower threshold when missing a potential clicker is costly and a higher one when wasted impressions are costly. The precision-recall curve supports this choice.
- **Skip time-of-day micro-optimization.** The temporal features add little, so scheduling effort is better spent elsewhere.
- **Communicate transparently.** Odds ratios give account managers a plain-language explanation of who the model favors and why.

11. Limitations and next steps

- **Very high scores on a small test set.** A 200-user test set means one error moves accuracy by half a percentage point. Results near 98% should be confirmed with stratified k-fold cross-validation and confidence intervals.
- **Idealized data.** A 50% click rate and this degree of separation are much cleaner than real campaigns, where CTR is often only a few percent. On live data, expect lower scores and a class-imbalance problem that will need attention (class weights, PR-AUC, threshold tuning).
- **Discarded information.** Country, City and Ad Topic Line were excluded. Testing encoded versions, such as target or frequency encoding, could reveal extra signal.
- **Linear assumption.** Logistic regression assumes linear effects on the log-odds. Comparing against random forest or gradient boosting would show whether non-linearities add value.
- **Correlated predictors.** Time on site and internet usage overlap, so individual coefficients should be read with care.
- **Deployment.** Before rollout, run an A/B test of model-targeted versus current targeting, and monitor for drift.

12. Conclusion

The project shows that a straightforward, interpretable logistic regression can separate likely clickers from non-clickers with about 98% accuracy on this dataset. Behavioral features dominate the prediction, with demographic features adding meaningful support. Used to rank and select audiences, the model gives ConnectSphere Digital a data-driven route to reduce wasted spend, lift CTR and improve ROAS for its clients. The next step is to validate the results with cross-validation and a live A/B test before full deployment.

Project report: Predictive Modeling for Click-Through Rate Optimization at ConnectSphere Digital. Analysis performed in Python with pandas, scikit-learn, matplotlib and seaborn.